



Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies

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PROFESSIONAL PROFILE

M.Sc. student in Quantum Technologies at TU Braunschweig with a B.Sc. in Computer Engineering and an AI-focused specialization. Hands-on and curious across both software and hardware, with practical Python and MATLAB work in computer vision, sensor-data analysis, CI/testing, and hardware-near coursework. Academic work spans communication and coding systems, photonics, semiconductor/GaN technology, and quantum information and computing; experience includes industry-university engineering work, IT support, and supervised university teaching.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering

Oct 2024 - Present

Technische Universität Braunschweig, Germany | In progress (Internship and Thesis)

Selected coursework: Quantum Communication Networks; Information & Coding Theory with Computational Experiments; Network Information Theory; Nonlinear Photonics;

Semiconductor/GaN Technology; Quantum Information & Computing

Academic engagement: presentations on QKD satellite communication, state-dependent channel coding, Quantum Communication Networks, and semiconductor crystal defects; plasmonic-metasurface journal-club literature review

B.Sc. Computer Engineering (Professionalization subject: AI)

2018 - Feb 2024

Azad University, Mashhad Branch, Iran

Overall grade: 17.28/20 | Ranked 5th out of 131 students

Relevant focus: Artificial Intelligence; Embedded Systems; C/C++ programming; SQL and database coursework

SELECTED ACADEMIC TECHNICAL WORK

Coding Theory Computational Group Lab

MATLAB-based group work

- Analyzed cyclic-code structure using Hamming distance and correction radius, and examined recorded bit-error behavior versus SNR and related error-correction concepts.

Nonlinear Photonics Analytical Coursework

Eleven exercise sets

- Completed analytical work on dispersion, phase matching, harmonic generation, nonlinear optical effects, and fiber nonlinearities, including refractive-index and gain/threshold calculations.

Information & Coding Theory Graded Coursework

Five analytical homework sets

- Completed graded work on binary/block coding, prefix codes and Kraft sums, relative entropy, Markov processes, and typical sets.

TECHNICAL EXPERIENCE

Teaching Assistant / Theory of Formal Languages and Automata

Sep 2023 - Jan 2024

Islamic Azad University, Mashhad Branch

- Led weekly exercise and problem-solving sessions for approximately 20 students.
- Corrected and evaluated assignments and exams within the supervised course.

Industry-University Project Member - Pouyesh Scheme

Apr 2023 - Sep 2023

Iran Khodro context | University Co-op

- Participated through the university in a MATLAB computer-vision vehicle-detection project.
- Ranked 1st among 50 participants in the relevant Pouyesh context.
- Subsequently received an invitation to work from Iran Khodro.

IT Support Intern

Jul 2022 - Sep 2022

Islamic Azad University, Mashhad, Iran

- Troubleshoot and repaired desktop computer hardware.
- Operating-system and system setup for staff and students.
- Provided staff and student support for local-network.

TECHNICAL SKILLS

Programming Languages	Python; MATLAB; C++ (coursework); Embedded C / ATmega32 (coursework); SQL (coursework)
Software / data / ML Development	NumPy; pandas; scikit-learn; Streamlit; PyTorch inference integration
Engineering domains	Git; GitHub; GitLab CI; pytest; flake8; Poetry Information & Coding Theory; Communications / Quantum Communication; Nonlinear Photonics; Semiconductor/GaN Technology; Quantum Information & Computing foundations / AI Implementation and Fine tuning / Automation

SELECTED SOFTWARE ENGINEERING PROJECTS

Gas Classification Pipeline and Simulation Dashboard

Python | pandas | NumPy | scikit-learn | RandomForest | Streamlit

- Constructed statistical features from multi-sensor time windows and used a stratified train/test workflow for six-gas RandomForest classification with class-probability output.
- Integrated pandas, NumPy, scikit-learn, and joblib in an interactive Streamlit simulation and probability dashboard.

Day/Night Vehicle Detection

MATLAB | Computer Vision | Image Processing

- Contributed to road-intersection vehicle detection under daytime and nighttime conditions in the university-industry Pouyesh context.
- Used HSV-value histogram logic for day/night selection, with GMM-based daytime foreground detection, morphology/connected-region processing, and nighttime threshold/component logic.

Activity Recognition and CI Configuration

Python | GitLab CI | pytest | flake8 | Poetry

- Structured a sensor-activity workflow comparing RandomForest and 1D-CNN approaches.
- Configured Poetry dependency management and GitLab CI quality checks with flake8 and pytest.

MRI Segmentation Inference Dashboard

PyTorch | Streamlit

- Integrated pretrained U-Net++ and EfficientNet-B4 inference workflows with preprocessing, weight loading, execution, and Streamlit visualization of masks, overlays, and outputs.

Hybrid Image Generator

MATLAB | Image Processing

- Authored a MATLAB script combining spatial-frequency components using Gaussian low-pass filtering and high-pass residual processing.

ATmega32 Embedded Systems Coursework

Embedded C | keypad/LCD | GPIO | 7-segment display

- Built coursework logic for a keypad/password digital safe with LCD messaging and a digital clock with push-button input, multiplexed display, time rollover, and manual adjustment.

C++ Programming Coursework

Bachelor-level algorithmic exercises

- Completed multiple Bachelor-level programming and algorithmic exercises in C++.

CERTIFICATES

Languages	English - C1 overall; TOEFL iBT 95/120 (22 May 2024) German - A2, currently attending classes Persian – Native
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Training & certificates	Quantum Machine Learning (QML) - Arya Quanta Institute Machine Learning - Kaggle Jan 2025 Introduction to Programming Using Python - Kaggle Jan 2025 Machine Learning and Data Science Bootcamp - Islamic Azad University, Mashhad Branch 96 hours Sep 2023 Python Programming - Jahad Daneshgahi, Khorasan Razavi Branch 40 hours 2021
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